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Hydrogen Gas (H₂) Generation in Hermetically Sealed Oil Filled Power Transformers: Evaluating HSE Risk, Addressing Challenges & Deploying Mitigation Measures

Anup Kumar Khatua, Senior Electrical Engineer, ADNOC Offshore, Abu Dhabi, UAE; Shaikha Salem Aljneibi, Process Engineer / Team Leader /A, ADNOC Offshore, Abu Dhabi, UAE; Abdulla Ahmed Balfaqeeh, Manager-TS, ADNOC Offshore, Abu Dhabi, UAE

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Abstract

This paper investigates the abnormal increase in hydrogen (H₂) levels observed during pre-energization Dissolved Gas Analysis (DGA) of oil (MIDEL-7131) in four hermetically sealed Power Transformers at an ADNOC Offshore facility. These transformers have been stored in a warehouse for over 18 months prior to installation. DGA testing diagnosed H₂ concentrations exceeded allowable limit of 100 ppm, as defined by IEEE C57.104 & IEC 60599. Recorded H₂ levels ranged from 323ppm to 1141ppm, indicating significant potential risk to HSE, operational integrity, business continuity, and loss of production.

An integrated approach has been developed through discussions with key stakeholders, including the Original Equipment Manufacturer (OEM), EPC Contractor and Consultants. This collaborative effort led to root cause analyses and the identification of the most efficient and cost-effective methods for mitigating hydrogen accumulation in the newly manufactured and ready for installation transformers.

Furthermore, the proposed mitigation measures both short term and long term are discussed, including degassing and dehydration of oil. With a focus on addressing challenges and proposing effective mitigation strategies, this paper provides actionable insights for practicing engineers. The paper concludes with recommendations for long-term monitoring and preventive maintenance (PM) practices and enhancing the reliability and safety of electrical systems in Oil & Gas industry. Future work will focus on monitoring the effectiveness of the implemented short-term mitigation measures and exploring additional strategies for further risk reduction as needed.

The study also explores emerging technologies including Artificial Intelligence (AI) for managing hydrogen-related risks contributing to enhance transformer reliability, extend equipment lifecycle and safety in high-demand operational environments; ultimately helping to prevent catastrophic failures.

Introduction

Background

The four hermetically sealed power transformers are preserved in warehouse for more than 18 months prior to the installation at Offshore Platform. Transformers have not been energized except during factory inspection tests (FAT) and have not been exposed to high temperatures, electrical stress or ionization. Preenergization DGA test of oil revealed higher concentrations of H₂ and other gases in all four transformers.

This paper explores possible sources of H₂ generation and analysis of gases in oil-filled transformers. The associated risks: such as reduced dielectric strength and potential transformer explosions, are examined. Furthermore, both short-term and long-term mitigation measures, continuous monitoring, oil treatment methods (degassing, nitrogen blanketing), preventive maintenance (PM) and design improvements, are discussed.

Hydrogen (H₂) generation and accumulation in hermetically sealed oil-filled transformers is a critical operational and safety concern. Unlike free-breathing transformers, hermetically sealed designs prevent gas exchange with the environment, leading to the accumulation of hydrogen and other dissolved gases. Excessive hydrogen buildup is often an indicator of insulation degradation, partial discharge activity, or thermal faults, and in extreme cases, it may pose explosion hazards.

Objective

This paper aims to:

- i. Investigate Potential Sources of Hydrogen Gas (H₂)
- ii. Analyze Dissolved Gas Analysis (DGA) Report
- iii. Evaluate HSE Risks
- iv. Recommend Actionable Short-Term and Long-Term Mitigation Measures
- v. Propose Design and Maintenance Improvements

Transformers Construction

The 3.3kV/0.433kV, 50Hz oil-immersed transformers (2 × 2000 kVA and 2 × 1600 kVA) are hermetically sealed, corrugated units designed for outdoor installation in offshore marine environment. Transformers are compact and filled with synthetic ester MIDEAL-7131 oil, are fully tight with nitrogen (N₂) blanket on top of the tank.

Key Protection and monitoring features include:

- DGPT2 protection device
- Oil Temperature Indicator (OTI)
- Winding Temperature Indicator (WTI)
- Pressure Relief Device (PRD)

The transformer tank is constructed from welded and pressed steel, featuring:

- Corrugated side panels with a thickness of 1.25 mm
- Bottom plate thickness of 6 mm
- Top cover thickness of 8 mm

The winding insulation is classified as Class-A. These transformers comply with IEC standards and are designed for continuous operation in harsh offshore conditions with space constraints.



Figure 1—Transformer 'as Installed' Layout

Dissolved Gas Analysis Report

In accordance with IEC 60422:2024, Dissolved Gas Analysis (DGA) is a key diagnostic tool for assessing transformer health. Commonly monitored gases include hydrogen (H_2), methane (CH_4), ethylene (C_2H_4), acetylene (C_2H_2), carbon monoxide (CO), carbon dioxide (CO_2), and other light hydrocarbons.

DGA testing was conducted as a part of pre-energization and commissioning procedure. The test was performed in ADNOC approved laboratory^[6] using ASTM D3612 Method C, which involves extracting dissolved gases through a headspace sampler and analyzing them using Gas Chromatography (GC) for accurate identification and quantification.

DGA Test results of all four transformers are presented in [Figure-2](#). Test results show higher concentrations of H_2 in all four transformers.

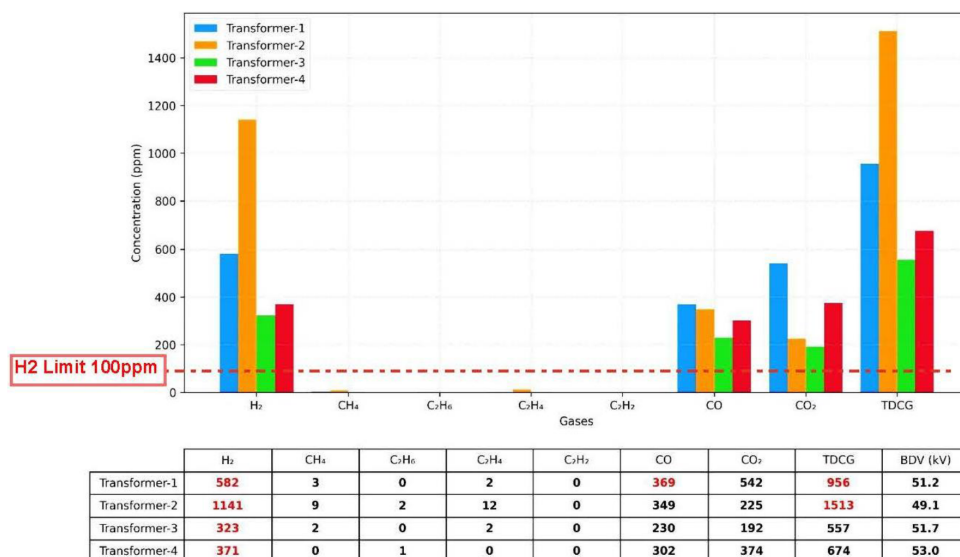


Figure 2—DGA Test Results

Potential Sources Of Hydrogen Gas (H₂)

Transformers were stored in the contractor yard at Abu Dhabi for more than 18 months before installation at site. Transformers were energized only during Factory Inspection Tests (FAT) and have not been exposed to high temperatures or direct sunlight. During FAT at OEMs works, temperature rise, Dielectric Tests (with Test Voltage 10kV on primary and 3kV on secondary windings) and Lighting Impulse tests have been performed. As per OEM, transformers are newly manufactured and have not been energized for long duration, it is unlikely that generation of H₂ is due to electrical stress or ionization. The source of H₂ may be due to chemical reactions and stray gas. Transformers are hermetically sealed, are fully oil tight with N₂ blanket on the top of the oil. Therefore, gas accumulated inside the transformers could not be released to the atmosphere.

However, referring to IEEE C57.155-2024, Clause 5.3, the predominant sources of H₂ gas are mainly due to Partial Discharge (corona) or Electrical Arcing.

It is noted from Figure-2; the H₂ values are higher than allowable limits for all four Transformers. TDCG (Total Dissolved Combustible Gas) values for Transformer-1 and 2 exceeded the allowable limit (Refer Table-1). However, BDV values are satisfactory.

Table 1—Typical Gas Concentration Limits In Transformer Oil Analysis

GAS / TEST SAMPLE	TYPICAL LIMITS
H ₂	100ppm
CH ₄	120ppm
C ₂ H ₆	65ppm
C ₂ H ₄	50ppm
C ₂ H ₂	1ppm
CO	350ppm
CO ₂	2500ppm
TDCG	720ppm
BDV	30kV

Typical gases that appear in transformers are hydrogen (H_2), methane (CH_4), ethane (C_2H_6), ethylene (C_2H_4), and acetylene (C_2H_2). These gases begin to form at specific temperatures and dissolve within the insulation oil of power transformer, as shown in Figure-3 and 4. The types and quantities of the gases that form will depend on the nature and intensity of the fault.

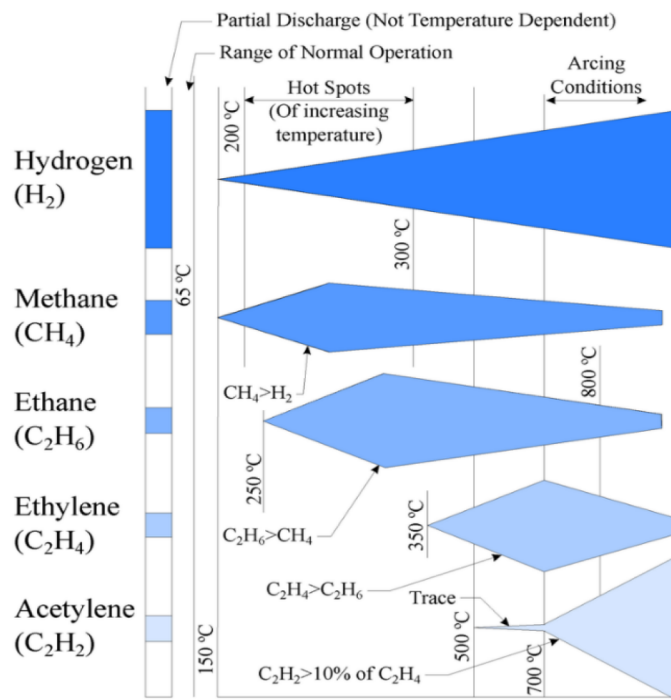
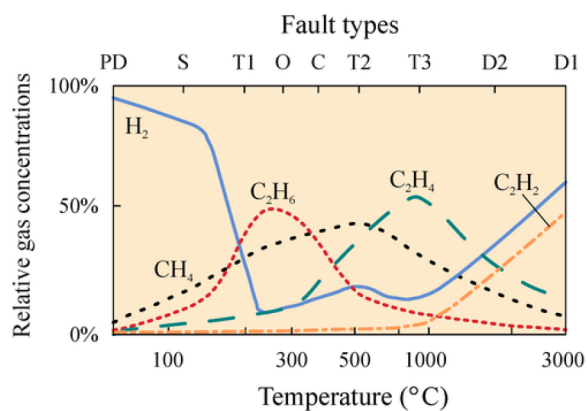


Figure 3—Gas Generation Chart



PD: partial discharges
D1: discharges of low energy
D2: discharges of high energy
T1: thermal faults (<300 °C)
T2: thermal faults (300–700 °C)
T3: thermal faults (>700 °C)
S: stray gassing
O: overheating of paper or mineral oil
C: possible carbonization of paper

Figure 4—Gas Concentration Profiles By Fault Type & Temperature

Hydrogen and methane begin to form in lesser amounts around 150°C. The production of hydrogen continues to increase as the temperature increases. At about 250°C, the production of ethane starts. At about 350°C, the production of ethylene starts. After reaching their maximum points, methane, ethane, and ethylene production goes down as the temperature increases.

Acetylene production begins between 500°C and 700°C. In the past, the presence of only trace amounts of acetylene was considered to indicate a temperature of at least 700°C had occurred, however, recent discoveries have led to the conclusion that a thermal fault (hot spot) of 500°C can produce trace amounts (a few ppm). Larger amounts of acetylene can only be produced above 700°C by internal arcing.

Between 200°C and 300°C, the production of methane exceeds that of hydrogen. Starting at about 275°C and up, the production of ethane exceeds that of methane. At about 450°C, hydrogen production exceeds all other gases until about 750°C to 800°C, at which point a larger amount of acetylene is produced.

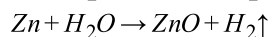
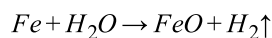
Thermal decomposition of cellulose materials will also begin at about 100°C or less and produce carbon monoxide (CO), carbon dioxide (CO₂), hydrogen (H₂), methane (CH₄), and oxygen (O₂). Therefore, it is imperative to operate transformers below 90°C.

The following possible sources of H₂ (both Fault Driven and Non-Fault Driven Gas) are shortlisted for these transformers:

1. Chemical Reaction between Synthetic Oil and Transformer Tank
2. Catalytic Hydrogen Generation from Oil Residues
3. Catalysis by Copper
4. Effect of Oil Inhibitors
5. Stray Gassing
6. Thermal Cycling / Faults
7. Partial Discharge (PD) / Corona Discharge (CD)
8. Free Water Electrolysis

Chemical Reaction between Synthetic Oil and Transformer Tank

Tank metals such as mild steel, galvanized steel, and certain stainless steels react with residual moisture and dissolved oxygen generates H₂ gas. This can occur during the manufacturing, transportation, preservation or commissioning stages, especially in environments with higher temperatures (Refer IEEE Publication – Magazine, 2006).



Catalytic Hydrogen Generation from Oil Residues

Another contributor to non-fault hydrogen formation is the catalytic decomposition of hydrocarbons, especially when residual mineral oil or varnish comes into contact with hot metal surfaces such as steel. Under high temperatures, such as during tank drying or post-assembly heating, hydrocarbons may undergo thermal cracking:



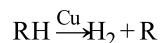
The phenomenon is accentuated when steel is used without proper passive action or when oil mist contacts heated metallic parts. In such cases, hydrogen production can occur even without direct electrical stress.

Catalysis by Copper

Copper components inside transformers can act as catalysts, accelerating the breakdown of oil. This catalytic action can promote gas formation even in the absence of high temperatures or fault conditions. The copper

surface encourages hydrocarbon degradation reactions, which lead to the generation of gases, particularly at lower temperatures as catalysts by definition lower the energy threshold of the reaction.

A simplified reaction catalyzed by copper:



Where RH represents a hydrocarbon molecule in the oil, and R is the remaining radical fragment.

Effect of Oil Inhibitors

Many transformer oils treated with inhibitors like butylated hydroxytoluene (BHT) to prevent oxidation. However, under certain conditions, the breakdown of these inhibitors themselves can contribute to stray gases. If the inhibitor concentration drops below optimal levels, the remaining hydrocarbons in the oil may become more susceptible to oxidation and breakdown.

Stray Gassing

Stray gas is a phenomenon where typical fault gases are formed when the transformer has not been exposed to high temperatures during loading. The most common stray gases are hydrogen, methane and ethane. Stray gassing refers to the evolution of hydrogen in the absence of thermal or electrical faults. These are chemical reactions in the presence of oxide films or paints/coatings containing metal-based drying agents. IEC 60599:2022 and ASTM D3612 both classify such occurrences as stray gas. Another common cause of stray gas is thermal cycling, where the transformer is subjected to repeated heating and cooling over its operating life. This cycling can cause minor breakdowns in the oil molecules without causing any obvious or severe fault in the transformer. Over time, these cycles lead to the accumulation of gases, without creating enough heat to burn off the gas or cause a more serious issue.

Thermal Cycling / Faults

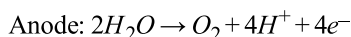
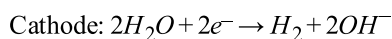
Thermal faults arise from overheating of transformer components, often due to overloading, cooling system failure, or contact resistance. Elevated temperatures break down hydrocarbon chains in the oil, releasing gases such as methane, ethylene, and hydrogen. The specific mix and concentration of gases depend on the temperature range and duration of overheating.

Partial Discharges (PD) / Corona Discharge (CD)

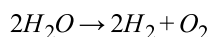
Partial discharges (PD) occur when localized dielectric breakdowns happen within the insulation, often due to voids, cracks, or contamination. These discharges do not completely bridge the insulation but cause enough stress to decompose oil molecules. One of the primary byproducts of PD is hydrogen. The intensity of PD correlates with the rate of H₂ generation, making it a reliable early indicator.

Free Water Electrolysis

Electrolysis of water is the process by which water (H₂O) is decomposed into oxygen (O₂) and hydrogen (H₂) gases when an electric current is passed through it. The reaction is:



Net Reaction:



However, since the transformers have not been energized for prolonged periods, the extend of hydrogen generation has not been considered this case.

Analysis Of DGA Report

DGA test was conducted in the ADNOC approved laboratory. It is noted from [Figure-2](#); the H₂ values are higher than allowable limits in all four Transformers. TDCG (Total Dissolved Combustible Gas) values for Transformer-1 & 2 exceeded permissible limits. However, Breakdown Voltage (BDV) values remain within acceptable limit, indicating no significant loss of dielectric strength.

The following analysis is based on the DGA results of Transformer-2 (most affected transformer).

Rogers Ratio Method (RRM)

Rogers Ratio is a diagnostic method used in DGA to help identify the type of fault occurring inside a transformer based on the ratios of specific gases. It's part of the IEEE C57.104 standard and is especially useful when acetylene is absent.

Ratio	Formula	Fault Type	Results	Remarks / Interpretation
R1	CH ₄ / H ₂	Low-energy discharge or corona	0.01	The low R1 and absence of acetylene suggest partial discharge rather than arcing.
R2	C ₂ H ₂ / C ₂ H ₄	Arcing	0.00	No acetylene, so no arcing
R5	C ₂ H ₄ / C ₂ H ₆	Thermal fault (temperature range)	6.00	The high R5 ratio points to thermal decomposition, likely of oil or insulation

Overall, the Rogers Ratios support the conclusion that the transformer is experiencing low-level thermal stress, not electrical fault.

Doernenburg Ratio Method (DRM)

Doernenburg Ratios are diagnostic tools used in DGA of transformer oil to identify fault types by comparing the relative concentrations of key gases such as hydrogen, methane, ethane, ethylene, and acetylene.

Ratio	Formula	Fault Type	Results	Remarks / Interpretation
R1	CH ₄ / H ₂	Low-energy discharge	0.01	The low R1 and absence of acetylene suggest partial discharge or low energy fault.
R2	C ₂ H ₂ /	Arcing	0.00	No acetylene, so no arcing
R3	C ₂ H ₄ / CH ₄	Thermal fault	1.33	The R3 ratio supports the presence of thermal decomposition, likely due to overheating of oil or insulation
R4	C ₂ H ₆ / C ₂ H ₂	Thermal fault	0.00	No ethane or acetylene, so not significant

Overall, the Doernenburg ratios confirm that the transformer is experiencing low-level thermal stress, not a critical fault.

Duval Triangle Method (DTM)

Duval Triangle is a graphical diagnostic tool used in transformer dissolved gas analysis (DGA) to identify fault types by plotting the relative percentages of methane (CH_4), ethylene (C_2H_4), and acetylene (C_2H_2) within a triangular diagram. Duval Triangle is drawn based on the DGA results (Refer [Figure-5](#)).

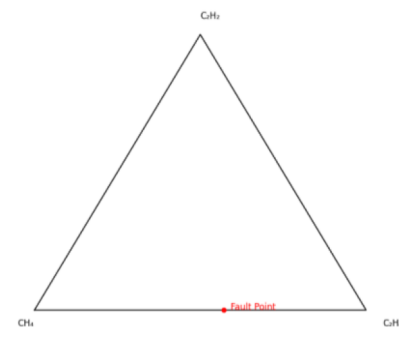


Figure 5—Duval Triangle

The fault point lies near the CH_4 - C_2H_4 axis, with no acetylene, indicating no arcing. This location typically corresponds to low-energy discharge or thermal fault, consistent with the Rogers and Doernenburg ratio interpretations.

Further Tan Delta (or Dissipation Factor) tests have been performed for all transformers; the measured Dissipation Factor / Power Factor values are within acceptable limits for synthetic ester oil ($\leq 1.0\%$), as per NETA ATS-2021 guideline, confirming good insulation health.

Therefore, it is generally concluded that the source of H_2 is mainly because of chemical reactions and stray gases (non-fault driven) not by the electrical stresses or ionization. Hydrogen (1141 ppm) and TDCG (885 ppm) are higher, suggesting accumulation of stray gases. Since the transformers are not energized except during FAT, partial discharge / prolong electrical stresses may be ruled out. No acetylene detected confirms no arcing or high-energy electrical faults. CO_2/CO ratio = 0.64 indicates cellulose insulation degradation is occurring.

How to Distinguish Stray Gassing from Fault Condition?

Since stray gases can produce gases commonly associated with fault condition as well, differentiating between the two is critical to ensuring the proper management of transformer health. It is concluded from Gas ratios, the H_2 gas produced through chemical reactions, i.e, non-fault driven not by thermal or electrical faults.

Evaluating HSE Risks & Challenges

Hydrogen (H_2) detected in hermetically sealed oil-filled transformers is dissolved in the insulation liquid, not in gaseous form. These transformers are fully sealed, preventing air ingress and eliminating the possibility of explosive gas mixtures under normal conditions. Any gas accumulation is directed to protection devices, which trigger alarms or trip signals if thresholds are exceeded.

Despite H_2 levels exceeded IEEE/IEC limits, no dielectric faults were observed during FAT and SAT testing, and DGA results confirm the oil's dielectric strength (BDV) remains intact. As per OEM, H_2 increase due to chemical reactions between zinc-coated tanks and transformer oil during prolonged storage; not for electrical faults.

While the dissolved H_2 poses no immediate threat to equipment or personnel, HSE concerns remain due to the potential for fault development. In rare cases, internal faults may generate oxygen, increasing

explosion risk. Therefore, post-commissioning monitoring—including daily logging of WTI, OTI, and pressure readings—is recommended, along with follow-up DGA testing.

Formal Risk Assessment (FRA)^[13] conducted to systematically identify potential hazards, analyze the likelihood and severity of associated risks, and implement control measures to minimize or eliminate those risks.

Hazards Associated with H₂ Accumulation:

- **Flammability and Explosion Risk**

Hydrogen is highly flammable and explosive at concentrations >4% in air. While hermetically sealed units are not air-filled, a slow leak or exposure to atmosphere during maintenance can form explosive mixtures.

- **Dielectric Degradation**

Hydrogen bubbles can reduce dielectric strength of the oil, especially at high voltage stress zones, increasing the risk of electrical breakdown.

- **Misleading DGA Interpretation**

In traditional dissolved gas analysis (DGA), H₂ is a leading indicator of PD or low-energy faults. In hermetically sealed systems, high H₂ without actual fault conditions may lead to unnecessary maintenance, misinterpretation.

- **Internal Pressure Build-up**

Gas generation in sealed tanks can raise internal pressure, stressing tank walls and gaskets. This may lead to bulging or tank rupture in extreme cases, especially in tropical climates where oil expands thermally.

As per OEM, the H₂ molecules are much smaller, they will gradually escape through the gaskets. After years, when the involved particles are consumed, the chemical process gradually slows down and H₂ level starts to stabilize or decline.

Short-Term Mitigation Measures

As the transformers are installed at site and ready for energization, any delay in commissioning impacts operations and results in production loss. Therefore, short-term quick-fix mitigation measures are recommended. The following short-term mitigation measures are suggested:

On-Site Oil Filtration for Dehydration & Degassing

Transformer oil filtration at site is a challenging procedure, specially at offshore platforms. The objective is to remove moisture and dissolve gases from the insulation fluid to restore dielectric integrity. The following process is followed (Refer [Figure-6 & 7](#)) as per Method Statement submitted by OEM:

- **Oil Heating:** Heat the oil to reduce viscosity and enhance moisture removal. The surface temperature of heater elements must be controlled to prevent fluid degradation. Recommended thermal power density is $\leq 0.8 \text{ W/cm}^2$.
- **Material Compatibility:** Ensure hoses and gaskets used in the fluid conditioning system are compatible with synthetic ester MIDEL 7131 oil.
- **Vacuum Treatment:** Apply vacuum to extract dissolved gases and water vapor. Note that corrugated transformer tanks are not vacuum rated; maximum allowable pressure differential is $\pm 0.1 \text{ bar}$.
- **Fine Filtration:** Use high-efficiency filters to remove particulate contaminants.
- **Oil Circulation:** Circulate the conditioned oil through the transformer until the total volume has been cycled at least six times to ensure uniform treatment.

- **Pre-Energization Hold:** Maintain the transformer in a fully oil-filled condition for a minimum of 48 hours before energization.
- **Fluid Sampling:** Collect oil samples post-filtration for laboratory analysis, including DGA and BDV.

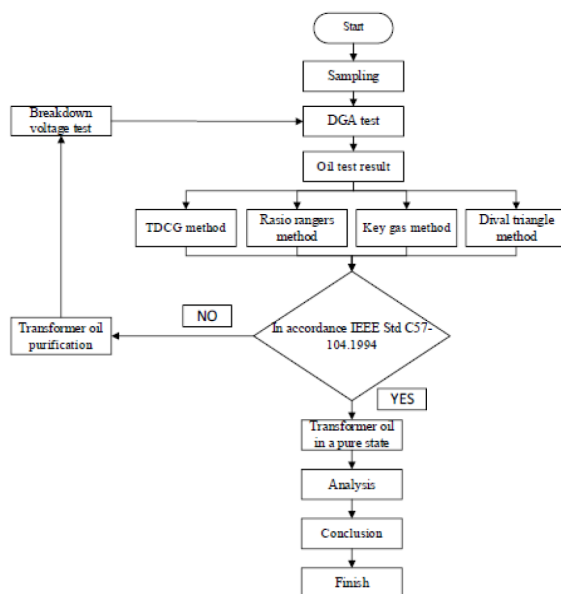


Figure 6—Flow Chart Of Filtration Of Oil

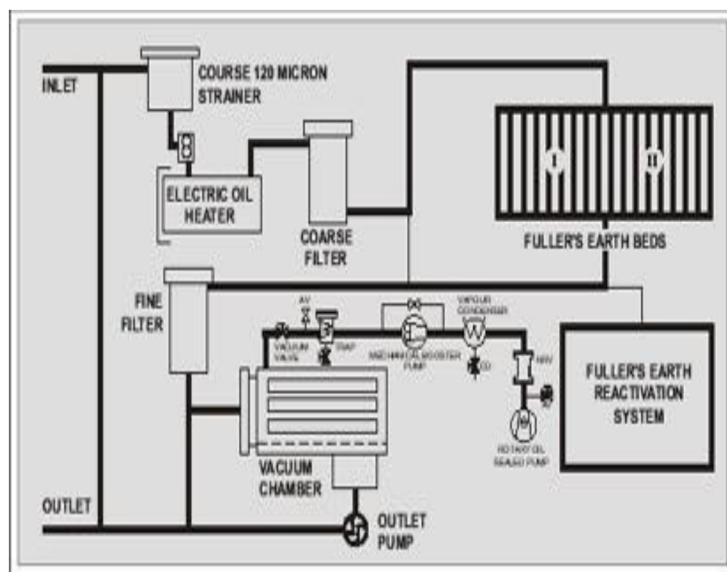


Figure 7—Oil Filtration Process

Outcome. Post-filtration DGA results confirmed hydrogen (H_2) levels were reduced to within IEEE/IEC allowable limits (<100 ppm) for all transformers (Refer Figure-8 & 9). Figure-10 illustrates the results for the most impacted unit Transformer-2.

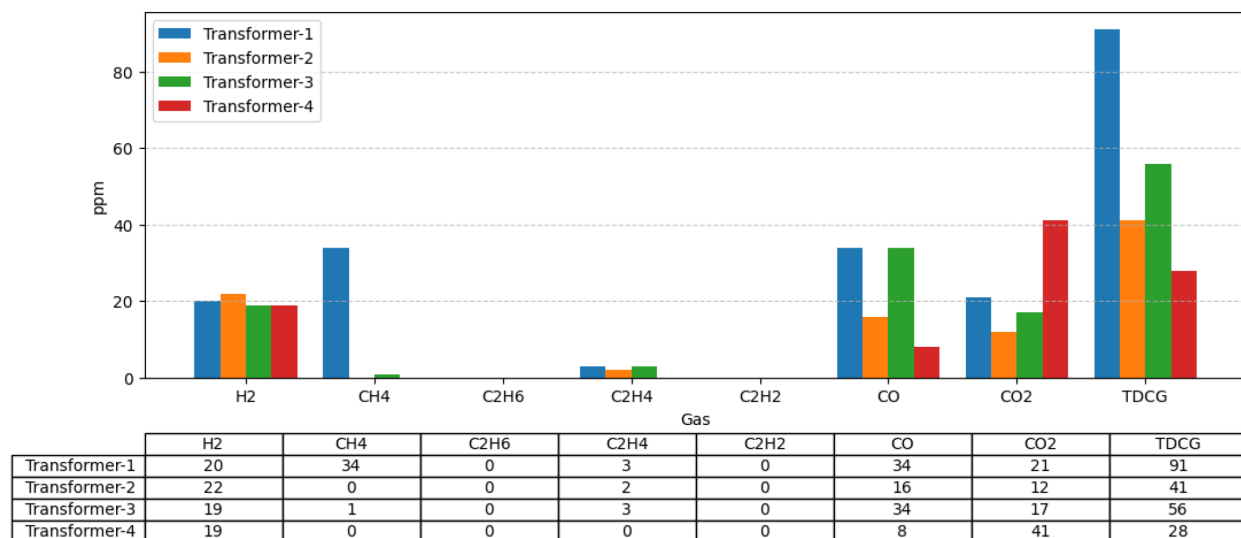
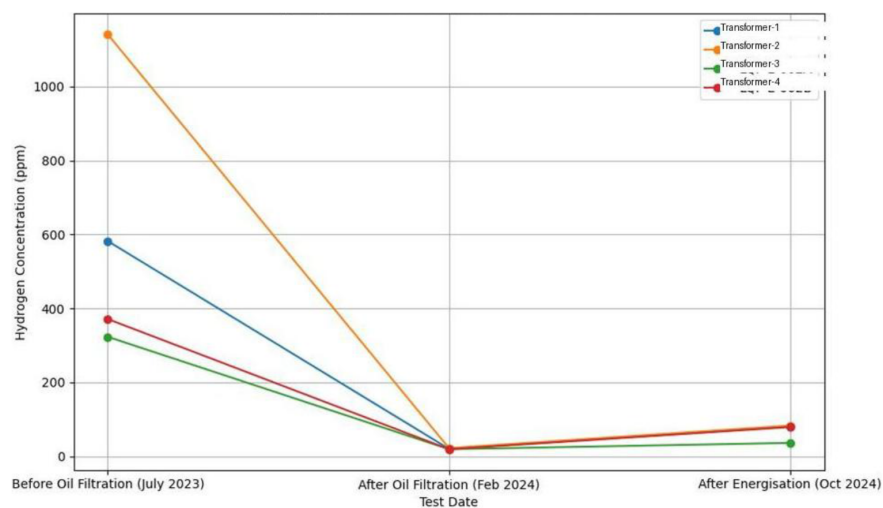
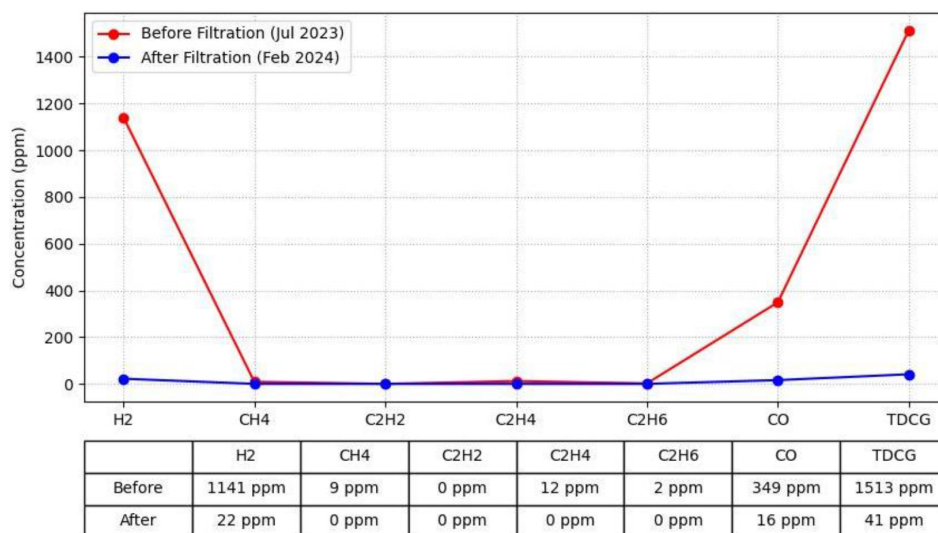


Figure 8—Dga Test Results Immediately After Oil Filtration

Figure 9—H₂ Concentration Before & After Oil Filtration And After 8 Months Of Operation On LoadFigure 10—H₂ Concentration Pre & Post Filtration For Transformer-2

Enhanced Monitoring via Annual DGA Testing

To ensure continued safe operation, implement annual DGA testing.

- **Trend Analysis:** If hydrogen levels increase without a corresponding rise in hydrocarbons (CH_4 , C_2H_2) or carbon oxides (CO , CO_2), the source is likely chemical gassing rather than thermal fault.
- **Continued Monitoring:** Maintain regular DGA testing to detect early signs of insulation degradation or fault development.

Preventive Maintenance (PM) Activities

Execute preventive maintenance in accordance with OEM recommendations:

- **Offline Testing:** Conduct partial discharge and tan delta testing if hydrogen levels continue to rise.
- **Thermal Imaging:** Use thermo-vision scanning to detect potential hot spots and localized heating.

Nitrogen Blanketing and Oxygen Suppression

Oxygen ingress can accelerate hydrogen formation. To mitigate this:

- **Dry Nitrogen Application:** Introduce dry nitrogen (dew point $< -40^\circ\text{C}$) at a positive pressure of 0.1-0.2 bar in the headspace.
- **Slow Purging Cycles:** Replace atmospheric air with nitrogen using slow purging cycles (#3 volume exchanges) to achieve $<2\%$ oxygen concentration.
- **Effectiveness:** This significantly slows oxidation kinetics and water-induced metal reactions, reduce hydrogen generation.

Post-Mitigation Monitoring and Warranty Extension

Following implementation of these measures:

- Transformers were successfully energized and are currently in service
- Hydrogen levels remain below 100 ppm after eight months of operation, posing no explosion hazard. However, H_2 levels have increased compared to those measured immediately after oil filtration.
- Annual DGA testing has been agreed upon with the contractor, and extended warranty coverage is in place

Long-Term Mitigation Measures

Hydrogen (H_2) levels continue to be monitored through Dissolved Gas Analysis (DGA) at one-year intervals. If the concentration of dissolved gases exceeds the permissible limits, the following long-term mitigation measures are recommended:

- **Oil Replacement:** Replace the transformer oil to remove accumulated gases and enhance dielectric properties.
- **Tank Replacement:** Consider replacing the transformer tank if internal chemical reactions persist due to material incompatibility or corrosion.
- **Installation of Online Gas Monitoring Systems:** Retrofit transformers with on-line gas monitoring systems to enable real-time diagnostics and enhance operational safety.

Long-term mitigation will be handled via extended warranty.

Conclusion

The investigation concluded that hydrogen gas (H_2) is generated mainly by chemical reactions within the transformer oil not by electrical stress or ionization. The transformers are newly manufactured and have not been energized, except during factory acceptance test (FAT), nor have they been exposed to high temperatures or direct sunlight.

These units are hermetically sealed and fully oil-tight, with a nitrogen (N_2) blanket above the oil surface. As a result, any gases formed inside the transformer are unable to escape, leading to accumulation and higher concentrations compared to free-breathing transformers. The most common gases observed are hydrogen, methane, and ethane. Hydrogen (H_2) concentrations in all transformers exceeded the allowable limits specified by IEEE and IEC standards.

Post-filtration, H_2 levels dropped below the IEEE/IEC threshold of 100 ppm, and dielectric parameters, such as breakdown voltage (BDV) and dissipation factor (tan delta), remained within acceptable limits. Further as per OEM, H_2 molecules, being the smallest and most diffusive, may gradually permeate through gaskets, and the rate of chemical reactions slowdown as reactive particles are consumed, leading to reduction of H_2 concentration.

After eight months of service of the transformers, DGA results indicated a slight increase in H_2 , but concentrations remain within safe limits. To ensure continued safe operation and without being caught off guard by an unaccepted ' H_2 spike', annual DGA monitoring is recommended as a part of Preventive Maintenance. This practice facilitates early detection of abnormal gas trends and ensures hydrogen levels remain below critical thresholds, thereby ensuring long-term operational integrity of the transformers in offshore environments. Should H_2 concentrations exceed the allowable limits defined by IEEE and IEC standards, long-term mitigation measures shall be implemented to establish a permanent solution.

Future Of DGA Diagnosis

Historically, Dissolved Gas Analysis (DGA) diagnostic techniques have relied on fault gas concentrations or ratios derived from expert practical experience, rather than on rigorous quantitative evidence. In this paper, conventional diagnostic methods, such as Rogers, Doernenburg ratios, and Duval Triangles are used for the interpretation of Dissolved Gas Analysis (DGA) in transformers. With the recent availability of extensive DGA datasets, researchers are increasingly motivated to pursue alternative methods for interpreting DGA data. These emerging approaches include Artificial Intelligence (AI), Machine Learning (ML), Fuzzy Logic, and Artificial Neural Network (ANN) based techniques. A notable advancement is the development of AI model that integrates KGM, DRM, RRM, and DTM methodologies, offering promising insights as illustrated in [Figure 11](#). Compared to traditional techniques used in isolation, AI-based models deliver more accurate and reliable transformer fault diagnoses. However, even though a majority of the AI approaches can diagnose faults with high accuracy, some of them fail to distinguish between fault driven and non-fault driven gases, so engineering judgment is still required.

By adopting AI, engineers can focus their expertise where it matters most, ensuring the safe and reliable operation of critical power infrastructure.

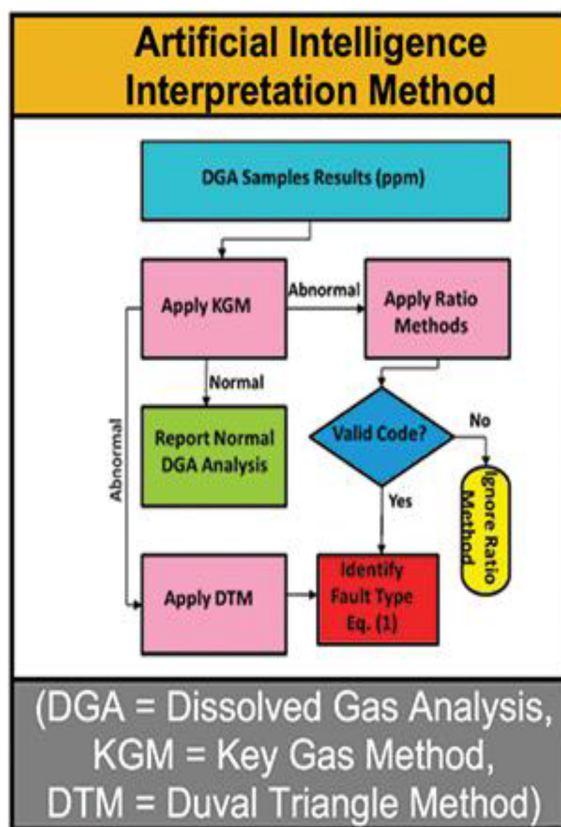


Figure 11—AI Models For DGA

Abbreviations

ADNOC : Abu Dhabi National Oil Company

ADNOC Offshore : An ADNOC Group of Companies involved in offshore oil production

AI : Artificial Intelligence

ANN : Artificial Neural Network

ATM : Atmospheric Pressure

ASTM : American Society for Testing and Materials

BDV : Breakdown Voltage

C₂H₂ : Acetylene

C₂H₄ : Ethylene

C₂H₆ : Ethane

CH₄ : Methane

CO : Carbon Monoxide

CO₂ : Carbon Dioxide

CD : Corona Discharge

CH₄ : Methane

CIGRE : Conseil International des Grands Reseaux Electriques

D : Electrical Fault

DGA : Dissolved Gas Analysis

DGPT : Detection of Gas, Pressure, and Temperature Relay

DTM : Duval Triangle Method

DRM : Doernenburg Ratio Method

EPC : Engineering, Procurement and Construction

E-HAZOP	: Electrical Hazard & Operability Study
ETCG	: Equivalent Total Combustible Gas
FAT	: Factory Inspection Test
FRA	: Formal Risk Assessment
GC	: Gas Chromatography
H ₂	: Hydrogen
HV	: High Voltage
HSE	: Health, Safety, and Environment
IEC	: International Electrotechnical Commission
IEEE	: Institute of Electrical and Electronics Engineers
KGM	: Key Gas Method
LEL	: Lower Explosive Limit
MIDEL 7131	: A synthetic ester dielectric fluid with high fire point
ML	: Machine Learning
NETA	: InterNational Electrical Testing Association
O ₂	: Oxygen
OEM	: Original Equipment Manufacture
ONAN	: Oil Natural Air Natural
OTI	: Oil Temperature Indicator
PD	: Partial Discharge
ppm	: Parts per million by volume of gas in oil
PM	: Preventive Maintenance
PRD	: Pressure Relief Device
PTW	: Permit to Work
RCA	: Root Cause Analysis
RRM	: Rogers Ratio Method
SAT	: Site Acceptance Test
SBRA	: Scenario Based Risk Assessment
T1	: Thermal Fault, $t < 300\text{ }^{\circ}\text{C}$
T2	: Thermal Fault, $300^{\circ}\text{C} < t < 700^{\circ}\text{C}$
T3	: Thermal Fault, $t > 700\text{ }^{\circ}\text{C}$
T	: Thermal Fault
TP	: Thermal Fault in paper
TDCG	: Total Dissolved Combustible Gas
TDHHG	: Total Dissolved High Heating Gases
WTI	: Winding Temperature Indicator

References

1. IEEE Electrical Insulation Magazine, January/February 2006 —Vol. **22**, No. 1, Case 2. 400 kVA, 12 kV – Distribution Transformer with a Zinc-Plated Tank, Manufactured in 1997, Hermetically Sealed.
2. OEM M/s Hitachi Energy Analysis report.
3. IEC60599, "Mineral oil-filled electrical equipment in service – Guidance on the interpretation of dissolved and free gases analysis," 2007.
4. IEEE C57.104, "IEEE guide for the interpretation of gases generated in oil-immersed transformer," IEEE Power Energy Society, 2019.

5. IEEE C57.155-2024 – IEEE Guide for Interpretation of Gases Generated in Natural Ester and Synthetic Ester-Immersed Transformers
6. Cigre-Technical-Brochure-771, "Advances in DGA interpretation," JWG D1/A2.47, 2019.
7. Transformer Oil Analysis Report # QMTL/R-279263/2023 Dtd. July 2023 by Qualitas Material Testing Laboratory, Abu Dhabi
8. Transformer Oil Analysis Report # 240131 Dtd. Feb 2024 by Wimpey Laboratories LLC, Abu Dhabi
9. Transformer Oil Analysis Report # 4512 Dtd. Oct 2024 by Bureau Veritas, Abu Dhabi
10. Transformer Oil Analysis Report #NIL Dtd. August 2023 by Hitachi Energy, Finland
11. ASTM D3612-96 Standard Test Method for Analysis of Gases Dissolved in Electrical Insulating Oil by Gas Chromatography
12. IEC 60422:2024 – Mineral insulating oils in electrical equipment – Supervision and maintenance guidance
13. UAD-RA-23-038: Formal Risk Assessment